

Answers

1. Define the Cartesian coordinate system in your own words.

Answer: *The Cartesian coordinate system is a method of representing positions in space using a set of numbers.*

2. Explain the purpose of the origin in the Cartesian coordinate system.

Answer: *The origin in the Cartesian coordinate system is the starting point or reference point for measuring positions. It is typically denoted as $(0, 0)$.*

3. Describe the role of the axis in the Cartesian coordinate system.

Answer: *The axis in the Cartesian coordinate system represents the reference lines along which positions are measured. In one-dimensional space, there is only one axis.*

4. On a number line, what do positive and negative numbers represent?

Answer: *Positive numbers on the number line represent positions to the right of the origin.*

5. What is the value halfway between 0 and 1 on the number line?

Answer: *The value halfway between 0 and 1 on the number line is 0.5.*

6. Give an example of a real-world application where the Cartesian coordinate system could be used.

Answer: *Examples of real-world applications of the Cartesian coordinate system include mapping locations, plotting graphs, and analyzing data in various fields.*

7. What is the significance of the origin in the context of the Cartesian coordinate system?

Answer: *The origin in the Cartesian coordinate system serves as the reference point from which positions are measured.*

8. Explain the difference between a point and a position vector in game development.

Answer: *In game development, the terms "point," "position," and "position vector" are often used interchangeably to refer to the same concept—the location of an object in space.*

9. Suppose you choose a position on the number line where the coordinate is -7. What does this represent in one-dimensional space?

Answer: *A position with a coordinate of -7 on the number line represents a point seven units to the left of the origin.*

10. How would you represent the point halfway between -5 and 2 on the number line?

Answer: *The point halfway between -5 and 2 on the number line would have a coordinate of -1.5.*

11. Calculate the coordinate of the point that is three units to the right of the origin on the number line.

Answer: *Three units to the right of the origin on the number line would result in a coordinate of 3.*

12. Can a position on the number line have a fractional coordinate? Explain why or why not.

Answer: *Yes, a position on the number line can have a fractional coordinate. We mostly mark the integer positions out of convenience, since it would be infeasible to mark every real value position.*

13. In game development, why is it important to accurately represent positions in space?

Answer: *Accurate representation of positions in space is important in game development to enable realistic movement, object interactions, and precise calculations.*

14. How would you represent the position of a player who is located two units to the left of the origin on the number line?

Answer: *A player located two units to the left of the origin on the number line would have a position with a coordinate of -2.*

15. Suppose you have two positions on the number line: -3 and 7. Calculate the distance between these two points.

Answer: *The distance between -3 and 7 on the number line is 10 units.*

16. Challenge: Given a position vector of -2.5, determine the position on the number line that is 1.5 units to the right of this vector.

Answer: *Given a position vector of -2.5, a point 1.5 units to the right of this vector would have a coordinate of -1.*